fintrust.

FinTrust Project Progress

Single living status and milestone record

RELEASE 1 | 13 October 2026

Source checkpoint risk-v2 / a474566

Authoritative consolidated documentation. Explicitly separates local verification, historical deployment evidence and pending live gates.

Fictional client: Imbokodo Bank. No real bank adoption or production compliance certification is claimed.

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1. Current Snapshot

LOCAL VERIFIED

FinTrust local candidate now includes Risk Model V2 and measurable component explanations.

Risk Model V2 and explainable UX VERIFIED: 903 PASSED | 0 FAILED | 1 EXPECTED SKIP (904 collected, +56 checks versus baseline). Final focused model/CSV/browser pass: 75; eight previously affected checks passed after explicit V2 oracle updates. Seven in-suite browser gates plus the separate four-role CSV/report gate passed. Preserved financial/audit data and credentials are unchanged. No historical rescoring, AWS contact, push or Linux run.

Risk model V2 uses one mutually exclusive amount component plus independent velocity, cross-border and possible-duplicate components. It remains deterministic, explainable and non-ML; its score prioritises investigation and is not a fraud or compliance verdict.

Amount <=50,000: 0; >50,000 to <=100,000: 15; >100,000 to <=200,000: 25; >200,000: 40. Velocity adds 25, cross-border adds 15 and possible duplicate adds 20. Maximum 100. LOW: 0-19; MEDIUM: 20-39; HIGH: 40-100. Thresholds use the original transaction currency; no FX conversion.

New assessments use fintrust-r2-rules-2026-10-13. V1 used overlapping HIGH_AMOUNT/VERY_HIGH_AMOUNT weights and additional transfer/merchant-marker signals. Existing V1 scores, classifications, source and rule version remain frozen; V2 does not rescore them. The current Release 1 validation, identity and review lifecycle remains in place.

Review eligibility migration 006 replaces the old hard-coded threshold with a version-aware guard. V1 LOW 30 remains ineligible; V2 MEDIUM 25 can open a review. The migration changes one existing trigger function, with no new tables or columns. Migrations 001-005 and historical rows remain unchanged. Live application remains a separate approved gate.

Controlled local presentation example: 150,000 ZAR; 8 accepted transactions in the closed 09:55-10:00 UTC window; country US versus ZA. Amount +25, velocity +25, cross-border +15, duplicate +0 = 65% - HIGH. Transaction -> Measured values -> Components -> Score -> Risk level -> Analyst investigation. This is not a probability of fraud; there is no invented partial velocity weight.

Baseline a474566 verified 847 passed / 0 failed / 1 expected skip, seven isolated browser gates and the preserved analyst01 workspace. That baseline is historical proof, not final proof of the changed V2 semantics. New focused, scenario, full-regression and browser evidence is recorded under docs/EVIDENCE/12-e2e/risk-model-v2/.

AWS untouched; nothing pushed; exact-source Linux verification and live AWS E2E remain pending. Behavioural dimensions and production-scale capacity remain planned.

2. Completed

LOCAL VERIFIED

Repository structure and source-path contracts; Analyst-first navigation and credential-preserving local restart; identity/deduplication and frozen risk rules; local review/account/audit workflows; authoritative append-only lifecycle and read APIs.

Historical base bootstrap/001/002 and service-role PG16 checks retained. Current lifecycle proof adds 21 PG16 tests. No live write is implied.

Named user governance, private local enrollment, active-role checks, immutable quarantine correction/outcome timeline, single-owner review claim/reassignment including escalated recovery, scoped queue filters and Admin/Analyst/Auditor browser controls.

Scoped Risk Event investigation, measured frozen rule evidence, same-currency account context, related transactions, filtered counts/sort, list-preserving detail, append-only Analyst notes, independent delivery state and compact user search. Existing quarantine/assignment controls remain verified.

Direct UI wording at 58fa83f, followed by README and authoritative documentation cleanup. Analyst workbench: six scoped metrics, active-review table, four evidence charts, issue links, independent errors and no fabricated SLA. Risk Events adds the existing Over SLA path; quarantine adds recorded age and a clear correction action; Admin user sorting/filtering and disable/reactivate remain verified.

Existing FinTrust-Monthly-Budget (USD 10/month; 50/80/100% ACTUAL and 100% FORECASTED) and My Zero-Spend Budget (ACTUAL > USD 0.01) verified, healthy and preserved. Same existing email subscriber on all five alerts; masked evidence retained. No duplicate/replacement budget created or current budget deleted.

13 October 2026 | Workflow continuity and own-upload correction locally verified: 847 regression passed, 1 expected administrator-only skip, zero failures; final four-role browser passed; three investigation/missing-evidence/own-upload browser checks passed. Implementation: 209700a. AWS untouched; no schema change, credential reset or push.

Local mock evidence/projections/alert claims survive restart; missing originals no longer hide case metadata/audit, and new correction fails closed. Field-based correction, worker outcome linkage, accepted-transaction and row lifecycle, compact quarantine queue, typed filters, human age and honest report periods are verified.

Approved own unknown-account correction uses immutable first authenticated upload authority and an existing assigned target. Duplicate upload cannot transfer ownership; known unassigned accounts and historical unowned cases remain denied. Admin/Auditor/Maintainer financial correction remains denied.

Running local server uses existing users/credentials and preserved financial/audit rows. No reset, seed, AWS contact or push.

Historical V1 gate: 13 October 2026 | Candidate source eb17ed5 includes risk-score display b73866c and the major console overhaul. Final candidate includes a test-isolation correction and a narrow report display/export correction. Full post-correction regression: 847 PASSED | 0 FAILED | 1 SKIPPED (848 collected); six in-suite browser checks passed against isolated PostgreSQL and mocked AWS. A separate four-role CSV/report browser gate also passed. Business rules, APIs and schema are unchanged.

Risk Model V2 and explainable UX VERIFIED: 903 PASSED | 0 FAILED | 1 EXPECTED SKIP (904 collected, +56 checks versus baseline). Final focused model/CSV/browser pass: 75; eight previously affected checks passed after explicit V2 oracle updates. Seven in-suite browser gates plus the separate four-role CSV/report gate passed. Preserved financial/audit data and credentials are unchanged. No historical rescoring, AWS contact, push or Linux run.

3. In Progress

LOCAL VERIFIED

AWS Integration remains In Progress. Local E2E is verified; live AWS E2E is pending. Budgets remain account-wide because Project and Environment cost-allocation tags are inactive. Recipient configuration is verified; activation status/actual delivery were not exposed or tested in this API check.

4. Blocked

PENDING VERIFICATION

Live promotion requires reviewed empty-database bootstrap and 001/002/003/004/005 chain, correct service grants and fresh exact-source Linux artifact gate. Existing fixed live runner does not include 003.

Cloud intake, FIFO/Streams mappings, authenticated API, SES delivery and operational monitoring still require controlled live evidence. Historical Linux artifacts remain valid only for their source.

5. Deferred

PLANNED

PLANNED - BEHAVIOUR-AWARE MONITORING: unusual amount against account baseline; unusual time/after-hours; unfamiliar country/geography or channel; new counterparty; dormant-account reactivation; bursts; rapid movement of funds; repeated payments below thresholds; historical behaviour deviation. None contributes points in Release 1.

A transaction at night is not automatically suspicious. It becomes a stronger signal when it differs from the account's normal behaviour. These features need adequate history, context, validation and governance before use.

CURRENT DESIGN: SQS FIFO buffers work; event-driven Lambda workers use per-account ordering and controlled concurrency; SQL owns the ledger and workflows, DynamoDB projections/delivery claims are independent, and retries/DLQ concern technical failures. The design has no NAT and remains cost-conscious. Local/mock proof is not proof of live wiring or production capacity.

FUTURE PRODUCTION HARDENING: 10k/100k/1M transaction load tests; concurrency/connection tuning; FIFO message-group strategy; high-throughput FIFO only if justified; RDS Proxy; queue-depth alarms; Multi-AZ; source-feed monitoring; configurable SLAs; alert aggregation; DR tests. The architecture is designed for elastic processing, but production-scale performance testing and connection tuning remain a future hardening step.

PLANNED - ADVANCED INTELLIGENCE: streaming ingestion, anomaly detection, AI Investigation Assistant with human review and authorised evidence, richer account/customer context, and governed adaptive thresholds. No AI or ML was added. Future features must improve What happened? Why is it risky? What should I do next?

6. Next Actions

PLANNED

Finish V2 local verification, commit this candidate, then perform a final local re-freeze. Use that new final source for exact-source Linux Python 3.12/x86_64 verification. Do not reuse the 847-test baseline as V2 proof.

After exact-source Linux approval: current packages/migrations, controlled AWS wiring, SES/retry/DLQ/report checks and live E2E. AI Investigation Assistant remains later. No live step was started in this refinement.

7. Milestone History

LOCAL VERIFIED

Selected major milestones from Git history and dated AWS evidence. Historical proof applies to its source; it is not rerun or promoted to the current candidate. Source paths resolve under docs/EVIDENCE where noted.

9e49a2f | Risk investigation implementation | 65 affected and 64 PG16 checks passed; complete regression 810 passed / 1 expected skip. Documentation/evidence commit follows tested source.

58fa83f | Direct UI copy | 18 focused tests passed, including both browser workflows. No behavior/API/database changes.

c35e41b | Analyst workbench | 19 focused, 82 affected and 811 regression tests passed / 1 expected skip. Three role/workbench browser gates and separate CSV scenario browser passed. Documentation/evidence follow this source checkpoint.

48399b9 | Local scenario verification | 20 scenarios, 125 selected tests passed; no product/schema changes. 13 October budget check | Existing monthly/zero-spend controls and masked recipients verified; account-wide scope recorded; no AWS mutations.

b73866c | Recorded transaction risk-score display; zero remains 0/100 and absent assessment remains Not assessed.

eb17ed5 | Major Analyst console overhaul: 111 affected checks and four final browser checks passed. Final complete regression is 847 passed / 0 failed / 1 expected administrator-only skip after the verification fixture isolation correction. Preserved-workspace smoke subsequently passed as analyst01; see final-local-freeze.json for the completed authentication gate.

Date	Commit / proof	Change / reason / validation
2026-09-29	cb729ea	docs: document FinTrust capstone design and presentation. Reason: Establish the capstone business and design baseline. Result: Historical design; not deployment evidence.
2026-09-29	2c98aa7	feat: implement Day 2 transaction data layer. Reason: Introduce the PostgreSQL transaction data layer. Result: Original implementation; current schema later extended.
2026-10-02	8d2cf05	docs: record budget-first dependency and network decisions. Reason: Avoid unmanaged standing cost and choose private service paths. Result: Budget-first decision recorded.

Date	Commit / proof	Change / reason / validation
2026-10-06	04276f9	Prepare isolated four-subnet Stage 1 network template. Reason: Isolate the Stage 1 private network. Result: Template preparation; later foundation evidence retained.
2026-10-08	3c53ab1	Record Stage 3 completion and Stage 4 costs and recovery. Reason: Record Stage 3 completion and next-stage cost/recovery gates. Result: Historical RDS deployment evidence; not current schema bootstrap.
2026-10-08	8745ad7	Record successful Stage 4 policy and template validation. Reason: Validate the isolated DynamoDB/SQS stage. Result: Policy/template validation recorded; later foundation verified.
2026-10-08	e14b6af	Add scoped legacy Logs tagging permission to canonical Stage 5 policy. Reason: Resolve provider-side tagged log creation authorization. Result: Scoped legacy tagging correction; diagnostics retained.
2026-10-08	e52ca2d	Prepare exact-qualified Stage 5 async permissions and rollback recovery. Reason: Separate qualified async configuration permissions. Result: Scoped fix and rollback procedure recorded.
2026-10-09	f118aa9	Record verified hardened Linux release and CI evidence. Reason: Close the earlier hardened release Linux gate. Result: Historical exact-source CI proof; not 7397f22 artifact proof.
2026-10-09	24ef12e	Define secure private RDS bootstrap through temporary Secrets Manager endpoint. Reason: Secure private master-credential access without NAT. Result: Temporary endpoint bootstrap design; later read-only proof.
2026-10-09	a136467	Record verified PostgreSQL 16 index helper and Linux runner artifacts. Reason: Prove narrow PostgreSQL 16 index-helper privileges. Result: PG16 helper and Linux runner evidence recorded.
2026-10-10	8fb2c6e	Record verified Release 1 lifecycle and exact Linux CI artifact. Reason: Bind dataset behavior and CI artifact to source. Result: Historical Release 1 lifecycle/package proof.
2026-10-11	906b555	Review authoritative Release 1 base schema and verification boundary. Reason: Detect the empty-live-database bootstrap gap. Result: Base source review; subsequent local proof retained.
2026-10-11	AWS evidence	Read-only private RDS preflight: secure TLS path and bounded read-only SQL. Result: missing base schema; no writes. Evidence: 05-rds/release1-live-readonly-alternative-preflight-result.md.
2026-10-11	AWS evidence	Temporary preflight cleanup to stop endpoint standing cost. Result: runner/network stacks, endpoint and temporary SG/ENIs removed; existing five stacks unchanged. Evidence: 05-rds/release1-temporary-preflight-cleanup-result.md.
2026-10-12	c6f4274	Use official AWS icons and surface the presentation ERD. Reason: Make architecture presentation accurate and readable. Result: Official AWS icons and ERD surfaced.
2026-10-12	9baae3f	Preserve existing local credentials when restarting Analyst workspace. Reason: Avoid replacing Analyst credentials on restart. Result: Credential-preserving local restart verified.
2026-10-12	a558051	Prioritize Analyst transactions and risk workflows. Reason: Prioritize Analyst work over presentation charts. Result: Transactions/risk workflow UX verified.
2026-10-12	6fbe773	Organize runtime source and audit FinTrust flow contracts. Reason: Make runtime boundaries clear without changing handlers. Result: 731 passed / 1 skipped; separate PG16 role proof.

Date	Commit / proof	Change / reason / validation
2026-10-12	7397f22	Wire authoritative transaction lifecycle audit and scoped API reads. Reason: Expose authoritative append-only lifecycle and scoped reads. Result: 756 passed / 1 skipped; focused 62; PG16 lifecycle 21.
2026-10-12	fb478fa	Named users, quarantine recovery and single-owner reviews. 789 regression passed / 1 administrator-only skip; 27 focused passed; 130 affected checks passed; 48 PostgreSQL 16 passed; browser passed. No AWS or push.
2026-10-12	636e4e5	Correct local identity-label encoding; explicit browser/source checks passed. Final regression: 789 regression passed / 1 administrator-only skip; 27 focused passed; 130 affected checks passed; 48 PostgreSQL 16 passed; browser passed.

8. Decision Log

LOCAL VERIFIED

Retain candidate-before-DB-validation chronology; keep processing, risk, review and delivery separate; accept all valid risk levels; original quarantine is terminal and corrections are new linked occurrences.

Retain >5 accepted transactions / 300 seconds, not >10/hour. Keep no-NAT/private RDS. Require in-memory bootstrap credential retrieval over a reviewed private endpoint and verify-full.

Master owns project explanation; Progress owns status/history; EVIDENCE owns proof. Technical contracts remain supporting sources, not alternative readiness declarations.

Reuse existing immutable correction/review history and current state models. Explicit Admin reassignment preserves escalated decision evidence. Keep SLA configurable without inventing a deadline; retain controlled local authentication and no role inheritance.

Datadog official investigation guidance informs information density and side-panel interactions only; the original FinTrust brief, RBAC and frozen risk rules remain authoritative. Reuse immutable review history for notes through additive 005; no second review model.

Reuse existing scoped read APIs for workbench counts. Display Not configured when no SLA exists; distinguish high-risk flags from delivery receipts; disclose bounded and overlapping counts. Keep human decisions, quarantine recovery and RBAC unchanged.

Preserve working budget names rather than create replacements. Budget thresholds are warnings, not a cap. Keep account-wide controls until activated cost tags and full attribution are reviewed; do not claim recipient delivery from subscriber configuration alone.

Approve Analyst correction of own unknown-account rows only: original authenticated first-upload receipt; corrected account must exist and be assigned; duplicate replay cannot confer ownership. No new schema or broader role authority.

Risk model V2 uses one mutually exclusive amount component plus independent velocity, cross-border and possible-duplicate components. It remains deterministic, explainable and non-ML; its score prioritises investigation and is not a fraud or compliance verdict.

Review eligibility migration 006 replaces the old hard-coded threshold with a version-aware guard. V1 LOW 30 remains ineligible; V2 MEDIUM 25 can open a review. The migration changes one existing trigger function, with no new tables or columns. Migrations 001-005 and historical rows remain unchanged. Live application remains a separate approved gate.

User-facing scores show a percentage, including 0%; missing assessment stays Not assessed. Rule-based risk score. This percentage represents the configured risk score, not the probability of fraud. Numeric API/storage semantics stay 0-100. Historical amount bands are grouped for display and scenario counts, with original identifiers available in technical evidence.

9. Issues and Resolutions

LOCAL VERIFIED

Tagged-create IAM failures: controlled diagnostics and provider-specific permission corrections recorded; static validation alone was insufficient.

Live database missing base schema: read-only preflight stopped incremental migration; temporary billable infrastructure cleaned up, evidence retained.

Stale local entry and login restart: authenticated workspace routing and existing hashed identity preservation fixed.

Lifecycle recovery: distinguish durable financial STORED from external projection failure; compare latest failure/recovery sequence so multiple failed attempts recover correctly.

Unresolved: current-source Linux gate, live bootstrap/003/004/005 privileges and integration. No claim of production readiness.

Observed Quarantine 503 isolated to lost in-memory original object versions after local restart. Durable ignored synthetic store added; exact version integrity checked; old missing evidence produces metadata/audit plus blocked writes. Browser checks were updated for actual CSV row numbering, governance-only Audit and newly approved mixed-batch visibility; final clean results supersede those failed intermediate checks.

Final gate found and corrected a test-isolation gap: Overview database reads did not use the disposable schema. Four focused failures were resolved by binding db.get_connection in the existing release_db fixture, then the entire suite passed with implicit developer dotenv loading disabled. The sole skip requires a disposable PostgreSQL administrator; prior separate PG16 role proof remains historical.

The separate CSV/report browser gate detected null risk-score fields being added to report rows by the shared presentation renderer. Risk projection is now limited to transaction and risk views. The final browser check verifies exported JSON equals backend rows and CSV values match the report response; no API, scoring or schema behavior changed.

Historical a474566 baseline: Authenticated preserved-workspace smoke PASSED as analyst01 (Risk/Compliance Analyst), using the existing credential entered privately in the browser. Overview, Transactions, Risk Events, Review Queue, Quarantine correction form, Accounts/Account History, Insights and Reports passed. Recorded scores, 0/100 and Not assessed are distinct; all ten Insights value tables match the API; report JSON matches backend rows. The investigation answers WHAT HAPPENED, WHY IS IT RISKY and NEXT STEPS.

Preserved financial and audit rows, credentials, roles, status and account assignments are unchanged. Normal sign-in metadata updated. No recovered case or owned unknown-account case exists in this Analyst scope; their recovery/authorization proofs remain separately labelled isolated evidence. No correction, claim, review outcome or user change was submitted for the smoke.

V2 revealed the database review guard hard-coded V1 >=40. Migration 006 retains versioned eligibility: V1 LOW 30 is denied; V2 MEDIUM 25 is eligible. No historical results are rewritten.

Frozen duplicate time proximity uses decimal text because Boto3 DynamoDB rejects float objects. UTC observed windows, unchanged CSV fixtures, explicit V2 expected scores and mobile/paginated browser navigation are verified.

Risk Model V2 and explainable UX VERIFIED: 903 PASSED | 0 FAILED | 1 EXPECTED SKIP (904 collected, +56 checks versus baseline). Final focused model/CSV/browser pass: 75; eight previously affected checks passed after explicit V2 oracle updates. Seven in-suite browser gates plus the separate four-role CSV/report gate passed. Preserved financial/audit data and credentials are unchanged. No historical rescoring, AWS contact, push or Linux run.

10. Evidence and Maintenance

LOCAL VERIFIED

Historical workbench proof: docs/EVIDENCE/12-e2e/analyst-workbench/verification-summary.json, regression/affected output and browser screenshots. Prior UI-copy, investigation, multi-user and AWS evidence remains unchanged.

Edit docs/source/project-progress.json and regenerate with python tools/build_project_docs.py. Keep historical results and explicit dates; do not overwrite evidence or mint another standalone status report.

Budget configuration: docs/EVIDENCE/01-budget/2026-10-13-budget-control-verification.json and masked settings evidence. Local scenarios: docs/EVIDENCE/12-e2e/scenario-verification/SCENARIO_RESULTS.md. Historical evidence remains unchanged.

Current UX proof: docs/EVIDENCE/12-e2e/analyst-ux-correction/verification-summary.json, isolated PostgreSQL regression results and authenticated browser screenshots. No AWS mutation, Linux gate, schema change or credential reset occurred in this phase. Current port-5001 server restart is verified in workflow-continuity evidence. Live AWS E2E remains pending.

Current proof: docs/EVIDENCE/12-e2e/workflow-continuity/. Preserve ignored build/local-runtime with the local database; it contains synthetic original evidence and mock delivery claims, not credentials. Historical counts remain dated. Local completeness does not prove live AWS readiness.

Final navigation/copy polish: 0056a59. Recovered transactions activate Transactions, not Quarantine; LOW entries use Risk Assessment. Three focused browser checks passed; a stalled investigation attempt was rerun alone and passed. Earlier full-suite result remains 847 passed / 1 expected skip; the final polish has targeted browser verification, not a second full-suite claim.

Historical a474566 freeze evidence remains under docs/EVIDENCE/12-e2e/console-overhaul/. Current V2 proof is docs/EVIDENCE/12-e2e/risk-model-v2/verification-summary.json; the containing commit is the reviewed model candidate for final local re-freeze.